

6. The US government monitors the consumption of different products. The table shows y , the amount of ice cream consumed, in millions of pounds, for x years since 2010.

x	0	1	2	3	4	5	6	7	8
y	4,354	4,222	4,114	4,030	3,970	3,934	3,922	3,934	3,970

The quadratic equation that models the amount of ice cream consumed, in millions of pounds, since 2010 is shown.

$$y = 12(x - 6)^2 + 3922$$

Determine when the amount of ice cream consumed in the United States would be 5,650 millions of pounds.

Work	$12(x - 6)^2 + 3922 = 5650$ $12(x - 6)^2 = 1728$ $(x - 6)^2 = 144$ $x - 6 = \pm 12$ $x - 6 = 12 \quad x - 6 = -12$ $x = 18 \quad x = -6$
Year(s)	2028